

ECON7040 Research Assignment

The COVID-19 pandemic affected the worldwide public health and caused negative economic shocks to the countries around the world. The pandemic caused problems in the public health area and negative economic shocks to Australia. To face the serious worldwide pandemic, Australian governments implemented strict public health policies. To respond to the negative economic shocks brought by the pandemic, the governments implemented expansionary monetary and fiscal policies. This report for ACID used the Australian macroeconomics data during the pandemic and after the pandemic. The New Keynesian partial sticky price model is used in the report to explain how the pandemic affects the real interest rate, inflation, output, employment and the real wage in macroeconomics.

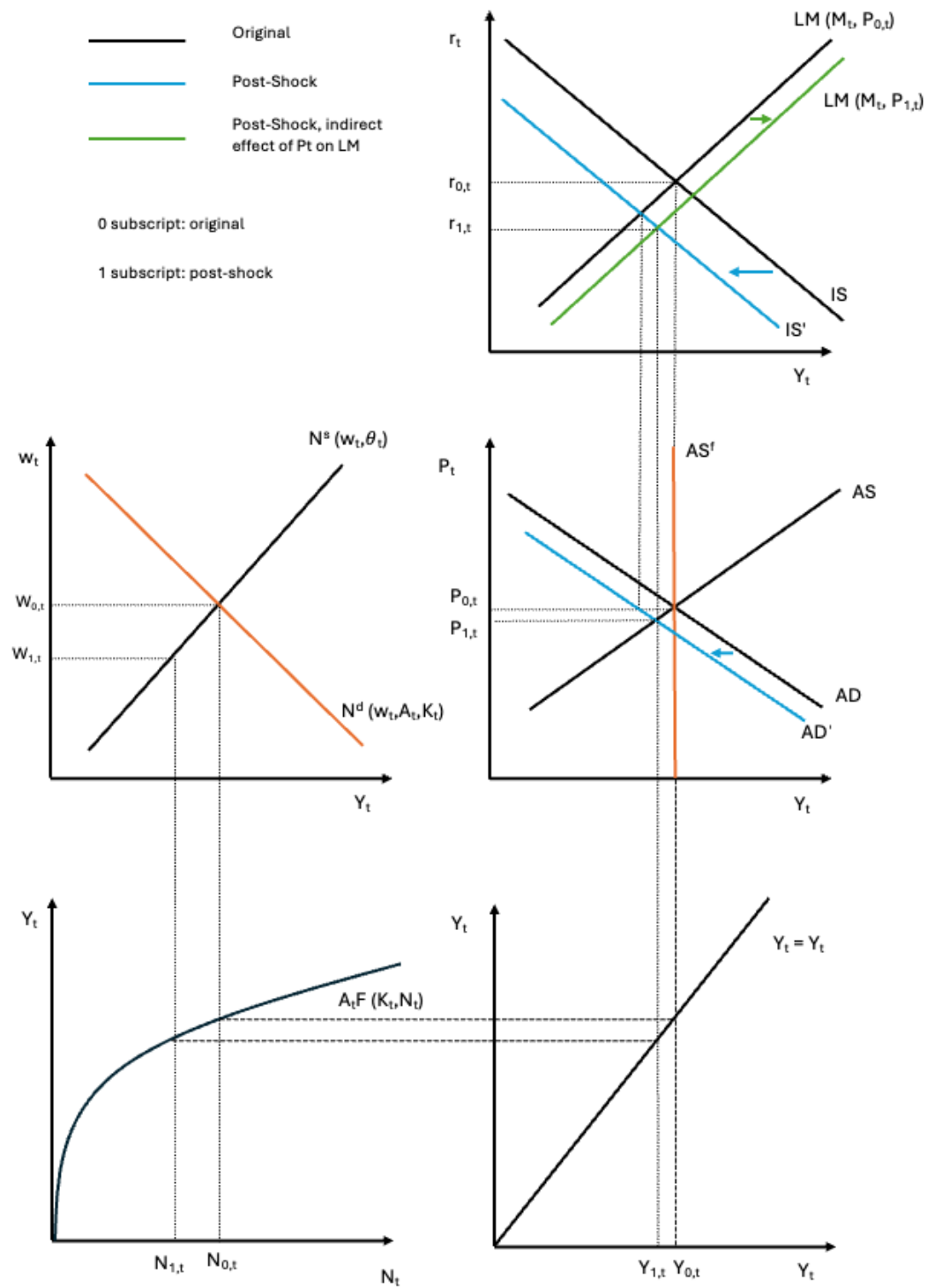
The Real GDP growth of Australia was -2% in 2020, 5.4% in 2021, 4.1% in 2022, 2.1% in 2023 and 1% in 2024 (International Monetary Fund, n.d.). According to the Reserve Bank of Australia (2025), the consumer price index Inflation was -0.3% in June 2020, increased to 3.8% in June 2021, 6.1% in June 2022, 6% in June 2023 and decreased to 2.4% in December 2024. The RBA set the CPI inflation target range is 2% to 3%. During the pandemic the lowest inflation is -0.3% in June 2020. After that, the inflation was still increased to the highest of 7.8% in December 2022. From 2023, the inflation started to decrease until 2.4% in March 2025. The RBA set the cash target rate at 0.25% from March 2020 to October 2020. From November 2020 to April 2022 the RBA set the cash target rate at 0.1%. From May 2022 to December 2024, the RBA continued to increase the cash target rate to over 4%. From November 2023, the cash target rate reached the highest 4.35%. The Australian Bureau of Statistics (2025) showed the real wage growth rate was 2.1% in June 2020, -2.0% in June 21, -4.4% in December 2022, 0.3% in June 2024. From June 2020 the real wage growth rate started to decrease until December 2023. The real wage growth rate was negative from June 2021 to September 2023. The lowest real wage growth rate was -4.4% in December 2022. From December 2023, it came back to positive. The Australian unemployment rate was 6.5% in 2020, 5% in 2021, 3.5% in 2022, 3.8% in 2023, 4% in 2024 and 2025. Employment to population ratio was 61.8% in 2020, 63.3% in 2021, 64.4% in 2022, 2023 and 2024. At the beginning of the pandemic in 2020, the employment decreased from April, but it came back to the level of January from December. The employment increased from 2021 to 2025.

The International Monetary Fund (2021) stated the Australian economic recovery was quick from the negative economic shocks caused by the pandemic in 2020. The expansionary monetary and fiscal policies such as decreased interest rate and Jobkeeper program provided support to face the negative economic shocks. The output level had recovered to a better level than before the pandemic output level until the 2021. According to IMF (2022) the inflation was more than the target range, the government implemented the contractionary monetary policy to face the high inflation, tight labour market and positive output gap. With

the high inflation, interest rate raised and lower real wages the consumer confidence decreased to a low level. In 2023, the inflation was still higher than the target range, the positive output gap was smaller, the interest rate increased, and the real wage decreased. RBA used tight monetary policy to face the high inflation. The unemployment rate and labor productivity were at a lower level in history (IMF, 2023). IMF (2024) stated the GDP growth decreased under the tight monetary policy. The monetary policy was still tight in 2024. The interest rate kept at 4.35% from November 2023. The inflation started to decrease to the target range.

When the pandemic started to affect the Australian economy, it caused negative IS shock. In the New Keynesian partial sticky price model, the effect of negative IS shock is displayed in figure 1. The direct effect of shock shifts the IS curve to the left. This caused a lower level of output, and the AD curve shifts to the left. The output and price level both decreased. The LM curve shifts to the right because of the indirect effect of the lower price level. The LM curve wouldn't shift too much, and the output wouldn't increase. The real interest rate decreased in the equilibrium. Because the production function is not affected directly and the output is decreased, the labour input is lower. The real wage decreased because the labour is based on the labour supply curve. The pandemic also affected the consumption and investment decreased.

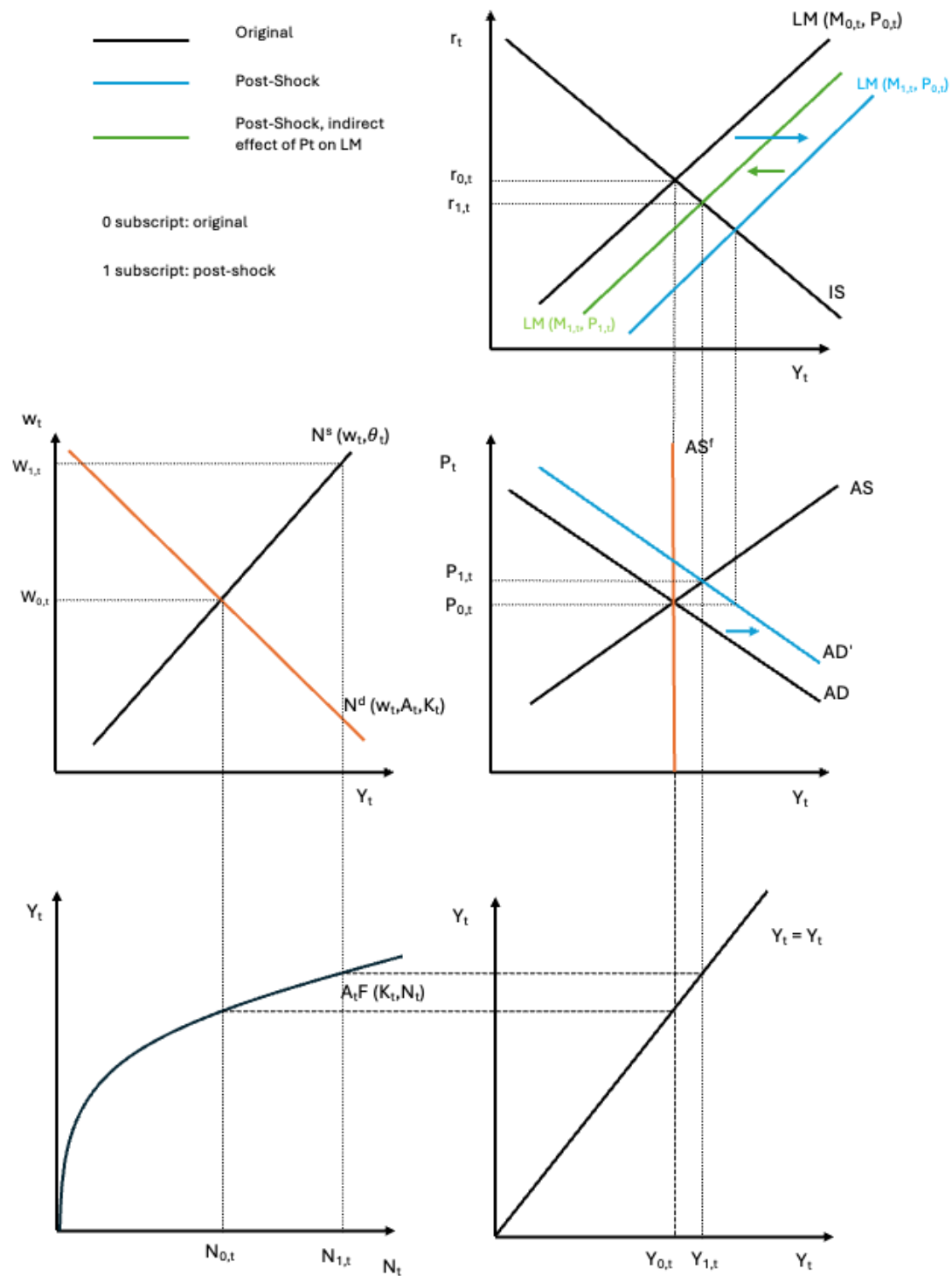
Figure 1



Effect of Negative IS Shock

Australia governments implemented expansionary monetary and fiscal policies to respond with the negative economic shocks caused by pandemic. The figure 2 shows the effect of loose monetary and fiscal policy in NK partial sticky price model. The loose macroeconomic policy increased money supply. The increased money supply directly affected the LM curve shifts right to the blue LM curve and led to the output increases. The AD curve shifts right to the AD' curve. The new price level and output is based on the point of intersection of AD' curve and AS curve. The AS curve is upward, both the price level and output increase. The increased price level affects the LM curve indirectly. LM curve moves to the left, the blue LM curve shifts left to green LM curve. The output was still increase and the real interest rate decreased. The production function didn't move, and because the output increased, the labour increased. The real wage increased with the higher level of output because the labour is based on the labour supply curve.

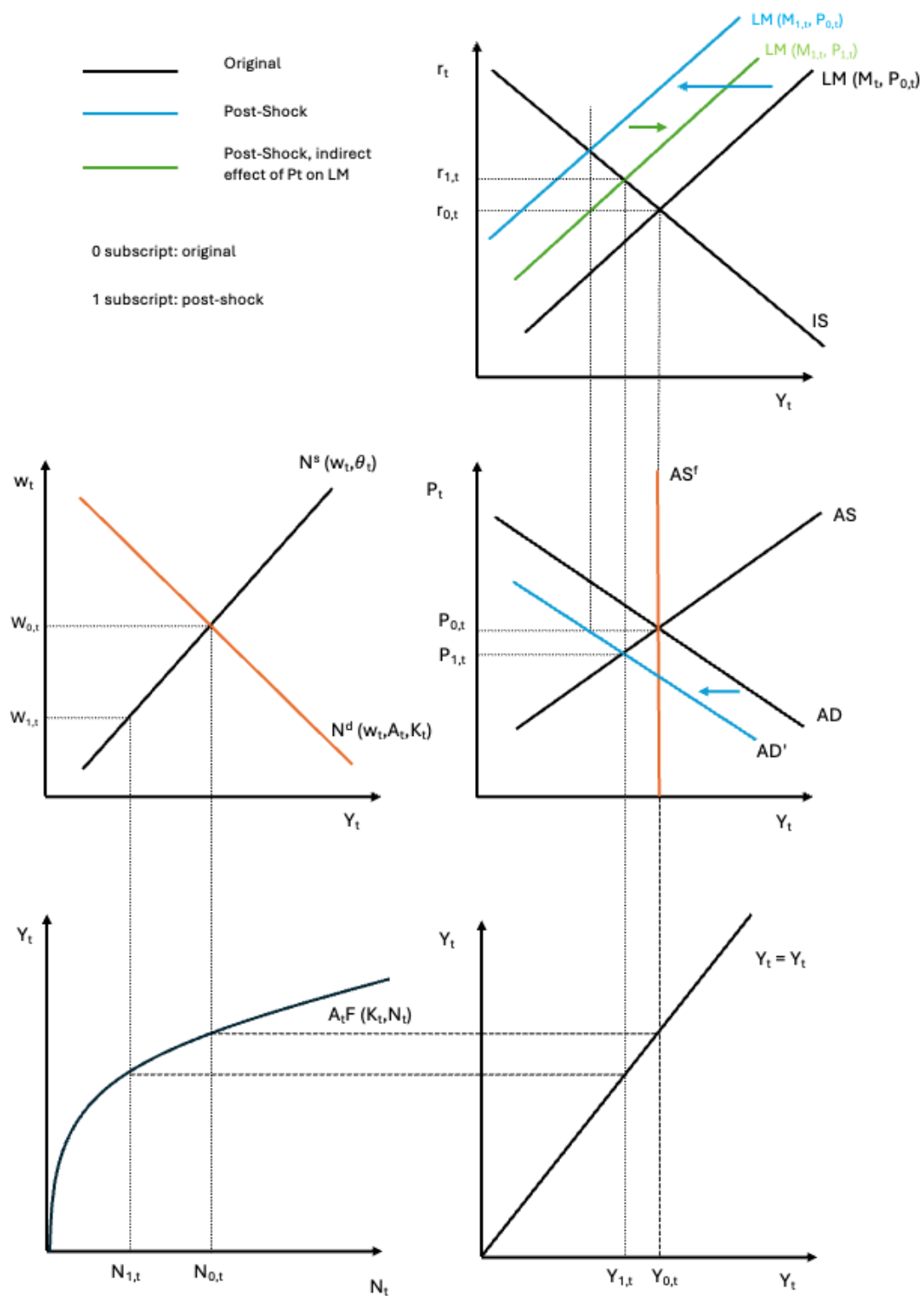
Figure 2



Effect of Expansionary Monetary and Fiscal Policy

Australia governments implemented contractionary monetary policy to face the high inflation. RBA increased the cash target rate. The figure 3 shows the effect of tight monetary policy in NK partial sticky price model. The tight monetary policy decreased money supply. The decreased money supply directly affected the LM curve moves to the left and the output decreased. The AD curve shifts to the left. Both the price level and output decreased. The lower price level affected the LM curve indirectly. The blue LM curve moves right to green LM curve. The real interest rate increased, and the output still decreased. The production function isn't affected, the labour decreased because of the decreased output. Decreased output caused the real wage decreased.

Figure 3



Effect of Contractionary Monetary Policy

The New Keynesian partial sticky price model explained how the pandemic affects output, inflation, the real interest rate, employment and the real wage. The explanation provided the economic framework of how Australia governments implemented macroeconomic policies and public health policies to face the shocks brought by the COVID-19 pandemic. The report can help Australia policy makers and corporate understand the pandemic caused economic impact better. The report aimed to help Australia policy makers and corporate prepare better to face the similar pandemic in the future.

Reference List

Australian Bureau of Statistics. (2025, May 15). *Labour Force, Australia*.

<https://www.abs.gov.au/statistics/labour/employment-and-unemployment/labour-force-australia/apr-2025>

Australian Bureau of Statistics. (n.d.). *Wages*. [https://www.abs.gov.au/statistics/measuring-](https://www.abs.gov.au/statistics/measuring-what-matters/measuring-what-matters-themes-and-indicators/prosperous/wages)

[what-matters/measuring-what-matters-themes-and-indicators/prosperous/wages](https://www.abs.gov.au/statistics/measuring-what-matters/measuring-what-matters-themes-and-indicators/prosperous/wages)

International Monetary Fund. (n.d.). *Australia and the IMF*.

<https://www.imf.org/en/Countries/AUS>

International Monetary Fund. (2021). *Australia: 2021 Article IV Consultation-Press Release;*

Staff Report; and Statement by the Executive Director for Australia.

[https://www.imf.org/en/Publications/CR/Issues/2021/12/02/Australia-2021-Article-](https://www.imf.org/en/Publications/CR/Issues/2021/12/02/Australia-2021-Article-IV-Consultation-Press-Release-Staff-Report-and-Statement-by-the-510754)

[IV-Consultation-Press-Release-Staff-Report-and-Statement-by-the-510754](https://www.imf.org/en/Publications/CR/Issues/2021/12/02/Australia-2021-Article-IV-Consultation-Press-Release-Staff-Report-and-Statement-by-the-510754)

International Monetary Fund. (2023). *Australia: 2022 Article IV Consultation-Press Release;*

and Staff Report.

[https://www.imf.org/en/Publications/CR/Issues/2023/01/26/Australia-2022-Article-](https://www.imf.org/en/Publications/CR/Issues/2023/01/26/Australia-2022-Article-IV-Consultation-Press-Release-and-Staff-Report-528629)

[IV-Consultation-Press-Release-and-Staff-Report-528629](https://www.imf.org/en/Publications/CR/Issues/2023/01/26/Australia-2022-Article-IV-Consultation-Press-Release-and-Staff-Report-528629)

International Monetary Fund. (2024). *Australia: 2023 Article IV Consultation-Press Release;*

Staff Report; and Statement by the Executive Director for Australia.

[https://www.imf.org/en/Publications/CR/Issues/2024/01/18/Australia-2023-Article-](https://www.imf.org/en/Publications/CR/Issues/2024/01/18/Australia-2023-Article-IV-Consultation-Press-Release-Staff-Report-and-Statement-by-the-543738)

[IV-Consultation-Press-Release-Staff-Report-and-Statement-by-the-543738](https://www.imf.org/en/Publications/CR/Issues/2024/01/18/Australia-2023-Article-IV-Consultation-Press-Release-Staff-Report-and-Statement-by-the-543738)

International Monetary Fund. (2024). *Australia: 2024 Article IV Consultation-Press Release; and Staff Report*.

<https://www.imf.org/en/Publications/CR/Issues/2024/12/20/Australia-2024-Article-IV-Consultation-Press-Release-and-Staff-Report-559811>

Reserve Bank of Australia. (n.d.). *Cash Rate Target*. <https://www.rba.gov.au/statistics/cash-rate/>

Reserve Bank of Australia. (2025, April 30). *Measures of Consumer Price Inflation*.

<https://www.rba.gov.au/inflation/measures-cpi.html>